Design and Implementation of a Natural Respiratory Rate Monitoring System Using Flex and Inertial Sensors on Medium-Power Microcontrollers

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Abstract. In the rapidly advancing realm of science and technology, the development of health monitoring systems is emerging as a vital trend, contributing significantly to improving human life quality. A noteworthy domain is the surveillance of individuals' natural respiratory rates during daily activities. These rates hold the potential to offer insights beyond simple health metrics and provide a comprehensive evaluation of overall well-being. This study proposes a wearable device system capable of autonomously tracking and recording natural respiratory rates during various activities, including standing, sitting, lying down, and walking. The system intelligently integrates flexible and inertial sensors. Flexible sensors detect body shape changes, while inertial sensors monitor movements and positions. This fusion enables the system to accurately and continuously measure natural respiratory rates, even with shifts in user positions and activities. Additionally, the system employs medium-power microcontrollers to process sensor data, conserving energy and ensuring real-time data processing. This approach guarantees the continuous and accurate update of respiratory rate information. With potential applications in healthcare, sports, and personal health management, this research promises numerous prospects. Monitoring natural respiratory rates through a system of flexible and inertial sensors on a medium-power microcontroller platform opens a new avenue for research, holding the potential to positively impact human health and overall quality of life.

Keywords: Respiratory · Human activity · Wearable · Flex sensor · Healthcare

1 Introduction

Recently, the development of flexible electronics and wearable devices [1–4] has created new opportunities for continuous health monitoring in daily life by utilizing sensors to support human health [5–8] surveillance, garnering significant attention. This has been

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achieved through the development and implementation of various flexible sensors. One of the significant challenges is integrating these sensors into wearable devices capable of tracking changes in the human body. A normal individual can provide a preliminary assessment of their health status based on some physiological information such as heart rate [9], respiratory rate [10], sweating [11], and so on. Particularly, certain health conditions like body temperature, stress, or survival risks can be easily detected through respiratory rate [12]. Respiration rate is a crucial indicator that not only reflects the body's respiratory condition but also can reveal early signs of various other health issues, including cardiovascular diseases, lung disorders, and psychological states. Respiration rate measurement solutions can be classified into two directions: i/Non-contact monitoring methods [13, 14] using motion, airflow through imaging, radar, and Wi-Fi; ii/Contact-based measurement solutions relying on sensors [1–4, 10] like accelerometers, thermal sensors, and more. The non-contact method may not be accurate, can be influenced by the surrounding environment, and may pose challenges when integrating into wearable devices. Furthermore, the use of cameras can potentially infringe on personal privacy rights. Using algorithms developed from previous research [15], and Butterworth data filtering, an integrated respiratory monitoring system on the wearable device was built with an accuracy of over 99%. With a different approach using accelerometer data, research [16] proposed an accelerometer sensor attached to the subject's chest to analyze respiration rate. The results achieved a deviation of 0.3%, which is quite impressive, but the study only focused on respiration rate investigation when the body is at rest, involving complex processing algorithms that are not suitable for real-time data. Furthermore, using respiration data measured by the accelerometer sensor requires high sensitivity and is susceptible to interference from basic body movements such as shivering and limb movements. Another study by Nabavi [17] established a similar wearable device to this research, with a respiratory rate (RR) measurement device in the form of a belt around the abdomen. Their measurement method achieved 95% accuracy by eliminating motion artifacts from the photoplethysmogram (PPG) signal, which adjusts the respiratory wave and provides a relatively accurate RR value. However, conducting surveys with multiple sensors attached to the fingertip for PPG measurement and a belt around the abdomen for RR measurement may be uncomfortable for extended periods and can lead to unintended data errors.

Fig. 1. The data collection in the form of a body garment is affixed to the bodies of volunteers.

Therefore, this study provides detailed information on the use of flexible sensors for respiratory rate measurement and presents the results directly on the wearable device (Fig. 1). The flexible sensor is integrated into a body-worn strap, and data is collected at a frequency of 40Hz, subsequently processed directly by the ESP-32S microcontroller.

Respiratory rate data is analyzed based on four basic activities, including standing, lying down, sitting, and walking, through the combination of flexible sensors and the accelerometer sensor ADXL345 on the wearable device.

2 Research Methods

2.1 Data Analytics

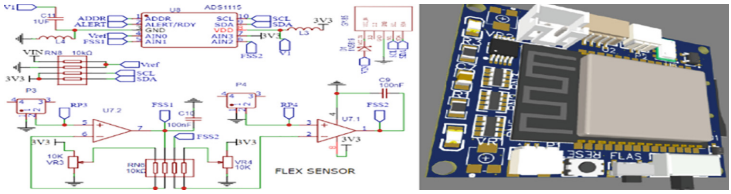

Fig. 2. Design of analog digital converter (ADC) and signal sensitivity adjustment

In this study, we designed a wearable device with an integrated ADXL345 accelerometer sensor, two 4.5-inch flexible curvature sensors, and a medium-performance microcontroller (ESP-32s) with memory limitations. The device was attached to the inguinal region (Fig. 2) and connected to a body garment with a flexible sensor. It simultaneously collected natural respiratory data at 40Hz frequency (f_s), corresponding to daily actions like lying, sitting, standing, and walking. Data records followed this structure: "*Timestamp, X, Y, Z-axis accelerations, left ADC value, right ADC value;*". The collected data were supported by volunteers aged 18–25, consisting of 5 males and 2 females. The volunteers adhered to common rules for wearing the devices and underwent pulse rate tests in various states. Respiration rate varies in amplitude, mean magnitude, and standard deviation continuously over time and depends on human posture (Fig. 3). To perform real-time respiration rate calculations, the respiratory rate data sequence is transformed into a standardized Z-score. This helps compare the data values to the mean and variability of the data. For a time-series respiratory rate data sequence $x[i]$ in the form $\{x_1, x_2, \cdots, x_n\}$ with $i = 1, 2, \cdots, n$ data points. The Z-score transformation is applied as follows:

$$z_i = \frac{x_i - \mu}{\sigma} \tag{1}$$

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{2}$$

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2} \tag{3}$$

With z_i is the Z-score standardized value of x_i. μ and σ represent the mean and standard deviation of the data sequence, respectively.

When a volunteer changes their posture, the amplitude range of the signal obtained from the sensor also changes (anomalies occur), such as transitioning from lying to sitting, or sitting to standing (Fig. 3). To identify anomalies in the data, a sliding window X_j with a size of 10 samples continuously slides over the respiratory rate data sequence in time. If the standard deviation of the window X_j exceeds the threshold τ or detects a change in the volunteer's actions, the standard deviation ($\sigma = 0$) will be reassigned at the next time point. According to the study [12], normal individuals breathe approximately 12–20 breaths per minute, which corresponds to 3 to 5 s/breath. With a sliding window size of 10, taking about 0.25 s, it does not significantly affect respiratory rate monitoring. Therefore, this data segment will be assigned the last value of the previous adjacent window ($z_{last_{X_{j-1}}}$).

$$z_{X_j} = z_{last_{X_{j-1}}} \text{ if } \sigma > \tau_{std_{X_j}} \; or \; and \; \sigma < \tau_{std_{X_j}}$$

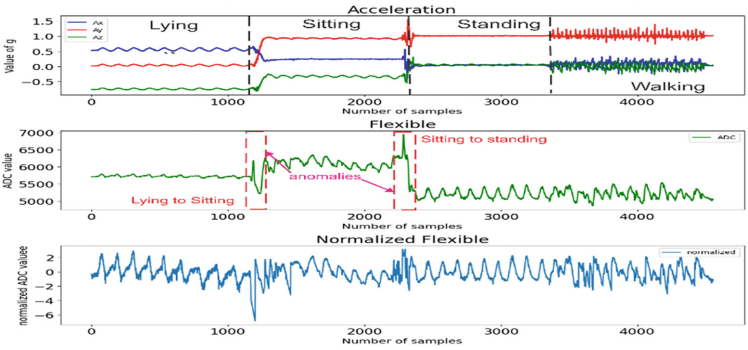

Fig. 3. Respiratory rate data corresponds to each activity

2.2 Noise Filter

To clean measurement data and remove noise, a second-order Butterworth filter is employed in this study. The low-pass Butterworth filter has a transform equation:

$$H(z) = \frac{B_z}{A_z} = \frac{b_0 + b_1 Z^{-1} + b_2 Z^{-2}}{1 + a_1 Z^{-1} + a_2 Z^{-2}} \tag{4}$$

where: $H(z)$ is the transfer function of the filter in the Z-domain. B_z is the transfer function of the feedforward system with coefficients b_0, b_1, b_2 corresponding to the filter's order. A_z is the transfer function of the feedback system with coefficients a_1, a_2 as the feedback filter's coefficients. These parameters are determined as follows:

Step 1: Determine the normalized frequency. In this study, $f_{lowcut} = 0.2$ Hz and $f_{highcut} = 0.5$ Hz.

Step 2: Determine the feedforward and feedback Parameters. Parameters for the low-pass filter.

Step 3: Apply the filter to the input signal time series $z[n]$ resulting in the filtered signal time series $y[n]$ in a general form:

$$y[n] = \sum_{i=0}^{2} b[i] \times z[n-i] - \sum_{j}^{2} a[j] \times y[n-j] \tag{5}$$

2.3 Respiratory Rate Determination Algorithm

In this study, we developed an algorithm with extended peak detection [15], flexible time, and adaptive Z-score calibration for different activities, combined with a Band Pass filter. This narrows down the value range and removes noisy frequency components from the signal. Particularly, the respiratory signal is enhanced with an Analog-to-Digital Converter using ADS1115 with 16-bit resolution to improve algorithm accuracy. Unlike step counting, the respiratory cycle tends to vary from 3 to 7 s depending on the activity and health status. Therefore, the respiratory rate calculation algorithm (Fig. 4) continuously collects and processes ADC data within a flexible time range from 0.5 to 7 s. This variability is achieved through true peak detection (breath cycle) and determining the breath cycle time from the true peak's left bottom to its right bottom (Fig. 6).

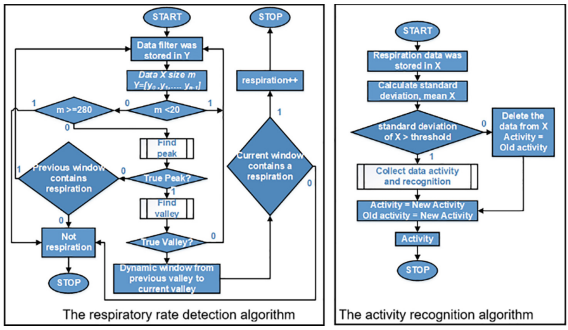

Fig. 4. The algorithm updates the ADC data related to respiration rate and detects respiratory cycles (left). If anomalies were detected in the respiration rate, activity was recognition (right).

2.4 Human Activity Recognition Model

Using machine learning on resource-constrained microcontrollers enhances the accuracy of human action recognition, posing a significant challenge [7, 8, 15]. This study focuses on investigating natural breathing rhythms in four states: sitting, lying down, standing, and walking. Consequently, our research continues to use the decision tree model with just two features: mean and standard deviation (Eqs. 2 and 3). Abnormal respiratory regions occur when volunteers transition between sitting, lying down, and standing. Continuous action identification during repetitive actions is unnecessary. To optimize energy efficiency, the system identifies actions only during static states (sitting, lying, standing) when abnormal respiratory regions are detected. Consequently, the wearable device's action recognition model activates only when there is a variation in respiratory rate, enhancing both performance and energy efficiency (Fig. 4).

3 Results

Figure 5 presents the statistical results of action recognition, showing that the system achieves a high level of accuracy, with actions in static states such as sitting, lying down, and standing being recognized with nearly 99% accuracy. Although there is a slight confusion when transitioning between standing and walking, it is not significant. This helps in evaluating the accuracy and objectivity of the respiratory signal state assessment.

Fig. 5. Confusion matrix for aggregating activity recognition results

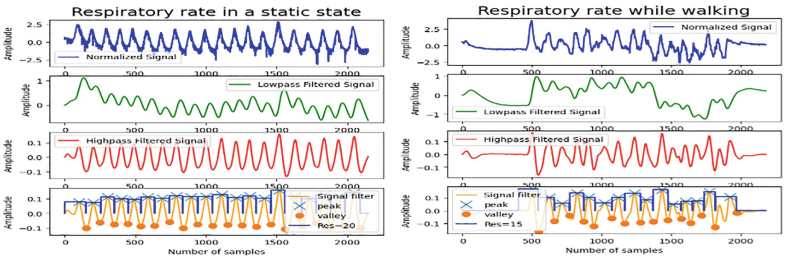

Fig. 6. Respiratory rate when the volunteer is in a static state (left) and in a moving state (right)

Table 1. Respiratory rate counting results for basic activities

Respiration activity	15	20	25	30	40
Lying	15	20	25	30	–
Sitting	15	20	25	30	–
Standing	15	20	25	30	–
Walking	16	18	24	31	–
Activities	–	–	–	–	42

The results of real-time respiratory rate counting achieved high accuracy in both motion and static states (Table 1). In the static state, the respiratory signal is noisy, tends to slope, and is evenly distributed over time, while in the motion state, the respiratory signal changes rapidly with the movement speed and stride. After passing through the Butterworth filter, the respiratory signal becomes smooth and clear. Statistical results

show a 100% detection rate for respiratory rate detection. This rate decreases slightly when a person is in motion, with a discrepancy in counts observed when performing 15, 20, 25, 30, and 40 breaths, ranging from 1–2 breaths. This discrepancy is negligible considering that the normal respiratory rate of a person is not fixed, and such an error is acceptable. Figure 6 illustrates that the natural respiratory signal of a normal person in a static state tends to be longer and more stable compared to when the person is in motion.

4 Discussion

This design aims to integrate smart devices into clothing for sleep monitoring, but placing it at the upper abdomen area can be inconvenient, especially when sleeping positions change. Furthermore, using an accelerometer sensor can be challenging for individuals with restless sleep patterns and unstable sleeping positions. In this study, a wearable device system integrated flexible sensors and accelerometer sensors to detect changes in abdominal muscle movements during breathing. These changes in skin surface tension and sensor resistance were recorded and analyzed using a 16-bit resolution ADC converter for four basic human activities. Various efforts have been undertaken to enhance the flexibility and portability of wearable devices. In a study by Park et al. [18], a novel wearable device was intro-duced utilizing a flexible capacitive pressure sensor. The sensor, coupled with a low-pass filter featuring a cut-off frequency of 20 Hz, effectively eliminated high-frequency noise. Its adaptable nature allows seamless integration into clothing and placement anywhere on the human body, making it a suitable op-tion for monitoring respiration signals. However, when compared to our system, this study falls short in terms of accuracy, displaying slight inaccuracies during human motion, particularly during exercise. Instances of increased peaks were observed within the same time period, especially when the subject engaged in physical activity, potentially affecting the precision of the recorded data in the presence of respiration abnormalities. Recognizing the scarcity of studies focused on measuring breathing rates during exercise, Yamamoto et al. [19] developed a non-invasive strain sensor capable of providing real-time respiration measurements during physical activity. The experi-mental results demonstrated a relatively high level of precision, with limits of agree-ment remaining within 3 breaths/min throughout the testing protocol. However, the study encountered challenges related to motion artifacts influencing phase detection, particularly at higher walking speeds. To address this, a numerical model was pro-posed to infer masked peaks under noisy conditions. In our work, we employ a low-pass Butterworth filter to effectively reduce noise while preserving crucial infor-mation, ensuring precise control accuracy and reliability.

5 Conclusion

The use of wearable systems to measure respiratory rate in various states often encounters issues with noise due to the presence of non-respiratory-related movements. Therefore, this study has developed an integrated system incorporating flexible sensors and acceleration sensors capable of monitoring respiratory rate with real-time data to address part

of this problem. The main strengths of the proposed research include: 1/ Utilizing an algorithm to directly measure respiratory rate on the wearable device, despite its limited performance, it has achieved over 99% accuracy in basic situations, even when the body is in motion; 2/ Employing an algorithm for automatic data calibration and detection of changes in unstable data, thereby enhancing the speed of action classification and energy savings; 3/ Designing an appropriately adjusted Butterworth filter for low-power control. In the future, we plan to further investigate the survival status of individuals and abnormal changes in respiratory and heart rate signals to develop a real-time, low-cost, and stable health monitoring system.

References

1. Balasubramaniyam, H., Vignesh, M.S., Florence S.E., Abanah, A.A.: Design and development of a IoT based flexible and wearable T-shirt for monitoring breathing rate. In: 2019 3rd International Conference on Computing Methodologies and Communication (ICCMC), pp. 376–379 (2019)
2. Jin, Y., et al.: Identifying human body states by using a flexible integrated sensor. npj Flex Electron. **4**(1), 28 (2020)
3. Presti, D.L. et al.: Wearable system based on flexible FBG for respiratory and cardiac monitoring. IEEE Sens. J. **19**(17), 7391–7398 (2019)
4. Di Tocco, J., et al.: A wearable system based on flexible sensors for unobtrusive respiratory monitoring in occupational settings. IEEE Sens. J. **21**(13), 14369–14378 (2021)
5. Thu, N.T., Dao, T., Bao, B.Q., Tran, D., Van Thanh, P., Tran, D.-T.: Real-time wearable-device based activity recognition using machine learning methods. Int. J. Comput. Digit. Syst. **12**(1), 321–333 (2022)
6. Nguyen, Q.H., Tran, D., Dao, T., Tran, D., Vu, T.T., Do, X.T.: A method of building a real-time human behavior recognition system based on machine learning algorithms. Res. Dev. Inf. Commun. Technol, **12/2022**(2), 83–88 (2023)
7. Dao, T.H., Yen, H.T.H., Hoang, V.N., Tran, D.T., Tran, D.N.: Human activity recognition system for moderate performance microcontroller using accelerometer data and random forest algorithm. EAI Endoresd Trans Ind. Networks Intell. Syst. **9**(4), 1–18 (2022)
8. Tran Thi Hong, N., Nguyen, G.L., Quang Huy, N., Viet Manh, D., Tran, D.-N., Tran, D.-T.: A low-cost real-time IoT human activity recognition system based on wearable sensor and the supervised learning algorithms. Meas. **218**, 113231 (2023)
9. Aygun, A., Ghasemzadeh, H., Jafari, R.: Robust interbeat interval and heart rate variability estimation method from various morphological features using wearable sensors. IEEE J. Biomed. Heal. Informatics **24**(8), 2238–2250 (2020)
10. Vu, H.-D., Tran, D.-N., Can, K.-L., Dao, T.-H., Pham, D.-D., Tran, D.-T.: Enhancing sleep postures classification by incorporating acceleration sensor and LSTM model. In: 2023 IEEE Statistical Signal Processing Workshop (SSP), pp. 661–665 (2023)
11. Cazalé, A., et al.: Physiological stress monitoring using sodium ion potentiometric microsensors for sweat analysis. Sensors Actuators B Chem. **225**, 1–9 (2016)
12. Rolfe, S.: The importance of respiratory rate monitoring. Br. Jounral Nurs. **28**(4), 504–508 (2019)
13. Massaroni, C., Schena, E., Silvestri, S., Taffoni, F., Merone, M.: Measurement system based on RBG camera signal for contactless breathing pattern and respiratory rate monitoring. In: 2018 IEEE International Symposium on Medical Measurements and Applications (MeMeA), pp. 1–6 (2018)

14. Lee, J., Yoo, S. K.: Radar-based detection of respiration rate with adaptive harmonic quefrency selection. Sensors **20**(6) (2020)
15. Dao, T.-H., Tran, D.-N., Hoang, Q.-T., Vu, H.-D., Huy, D. T., Tran, D.-T.: Developing real-time automatic step detection on a low-cost, performance-constrained microcontroller. In: 2023 IEEE Statistical Signal Processing Workshop (SSP), pp. 150–154 (2023)
16. Fekr, A.R., Janidarmian, M., Radecka, K., Zilic, Z.: A medical cloud-based platform for respiration rate measurement and hierarchical classification of breath disorders. Sens. **14**(6), 11204–11224 (2014)
17. Nabavi, S., Bhadra, S.: A robust fusion method for motion artifacts reduction in photoplethysmography signal. IEEE Trans. Instrum. Meas. **69**(12), 9599–9608 (2020)
18. Park, S.W., Das, P.S., Chhetry, A., Park, J.Y.: A flexible capacitive pressure sensor for wearable respiration monitoring system. IEEE Sens. J. **17**(20), 6558–6564 (2017)
19. Yamamoto, A., et al.: Monitoring respiratory rates with a wearable system using a stretchable strain sensor during moderate exercise. Med. Biol. Eng. Comput. **57**(12), 2741–2756 (2019)